Name	Helen Zoljargal
Student ID	33936897
Studio Session	Tuesday 12-2

Part A: URL of the Visualisation

https://helenzolja.github.io/FIT3179-Data-Visualisation-2/

Part B: Domain Description

Describe your domain chosen for DV II

The domain of my DV2 project is environmental exploration and science, as the visualisation explores Australia's plant biodiversity, focusing on native plants, distribution across Victoria and other status and characteristics of flora across the country.

The aim of the visualisation is to investigate how plant species are distributed across Australia, various plant characteristics, vegetation characteristics and environmental conditions that contribute to biodiversity patterns. The project draws on data from Atlas of Living Australia, iNaturalist and ANBG Australian flora statistics for raw data with other external sources referenced for context and images. Since the data is mostly occurrence records collected from citizen scientists, the visualisation reflects where and when people look for plants, and skew towards a "human bias" rather than scientific accuracy, which the page acknowledges directly, normalizing data where appropriate and necessary.

Describe the Why and Who aspects of your DV II

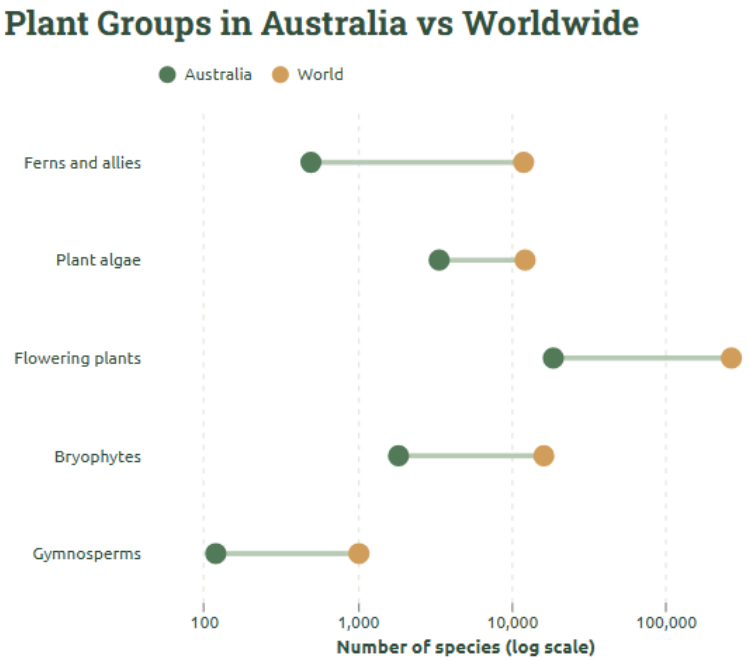
Why?

The goal of the visualisation is to help readers understand and explore Australia's distinctive flora, plant diversity, seasonal and regional trends and patterns, as well as why it needs protecting. Each section supports different concepts that tie the whole story together.

Who?

The primary audience is the interested public, ranging from nature lovers, hikers, gardeners to citizen scientists and students. The visualisation is not meant to be scientifically accurate but rather provide an accessible overview and contextual exploration of the data used. Hence, scientific terms such as species names are often referred to by its vernacular names, tooltips for exact values, and images and annotations littered throughout for visual understanding.

Chart 1 – Dumbbell chart



Note: The axis uses a log scale, so each gridline is ten times the one before.

Source: DCCEEW, Numbers of Living Species in Australia and the World (2009).

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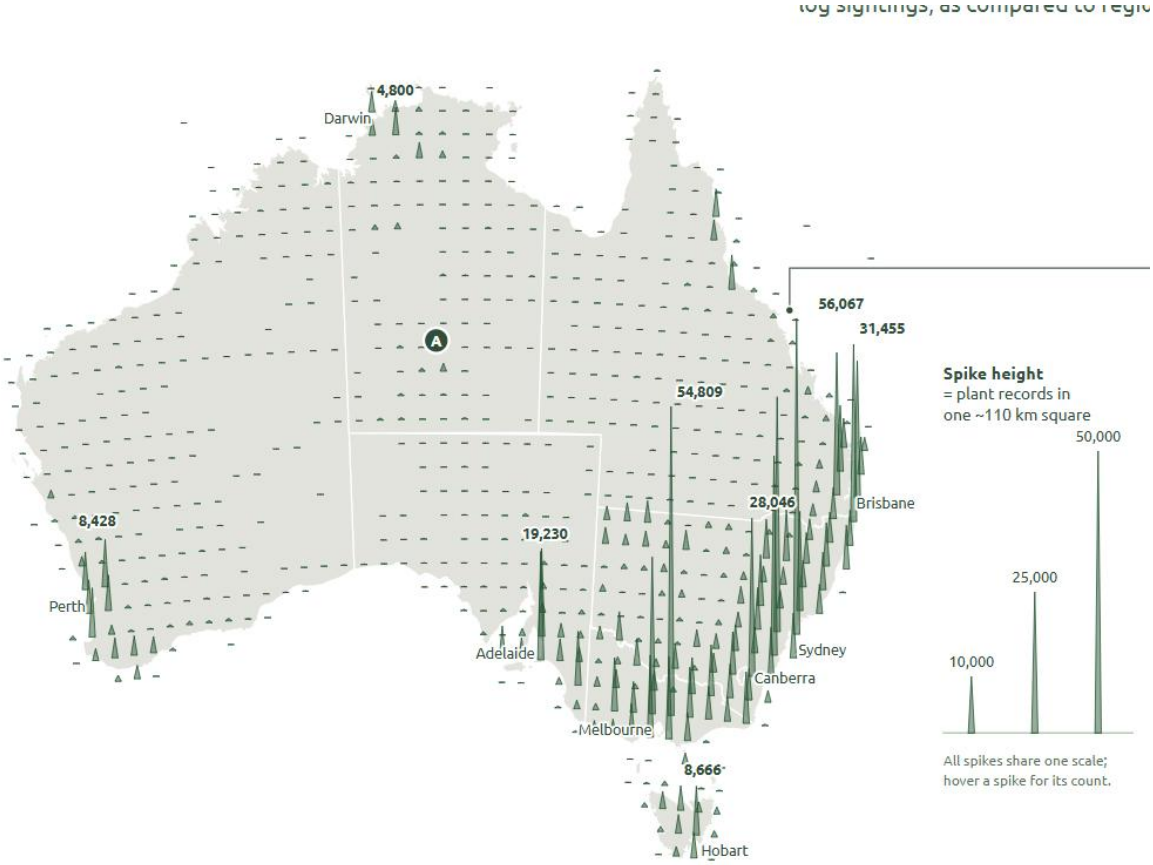
Why?

Chart 1 introduces the scale of Australia's plant diversity by comparing the number of described species in Australia with the world total for 5 major plant groups, using data taken from DCCEEW. Its main tasks are to compare Australian and worldwide species counts within each plant group and lets viewers summarise the overall pattern.

How?

A dumbbell chart uses paired circular marks connected by lines to represent Australian and worldwide species counts for each plant group. Horizontal position encodes species count on a logarithmic scale, while vertical position represents plant groups and colour hue distinguishes geographical categories Axis lines and tick marks are removed, and gridlines are kept faint to improve the data ink ratio while embedded tooltips let viewers look up the exact species count for each point for further investigation.

Chart 2 – Spike map



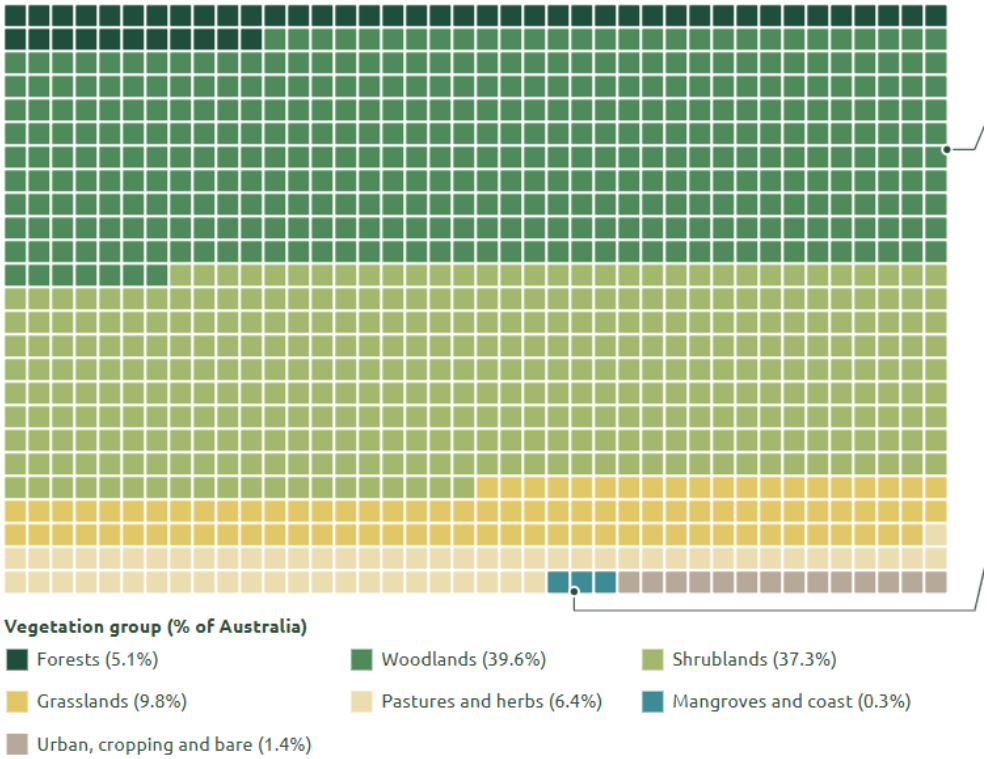
Why?

The graph shows where plants were recorded across Australia over the year of 2025, sources from ALA. Its main task is to locate where plant observations are concentrated and to identify the extremes. For example, the tallest spike near each capital city is labelled and annotates which allows viewers to compare observation levels between regions and discover the overall spatial pattern.

How?

A spike map is used because it shows a quantity at many locations while keeping the geographic context. Records are grouped into squares about 110km across, and each square is drawn as a spike rising from its centre on an equal area map of Australia. The position of the spikes encodes geographic location, and their height encodes the number of plant records in that square, a quantitative attribute. Compared to a proportional area map, spikes appear cleaner, more distinctive and allow the viewers to estimate any spike's value by comparing heights.

Chart 3 – Waffle Chart



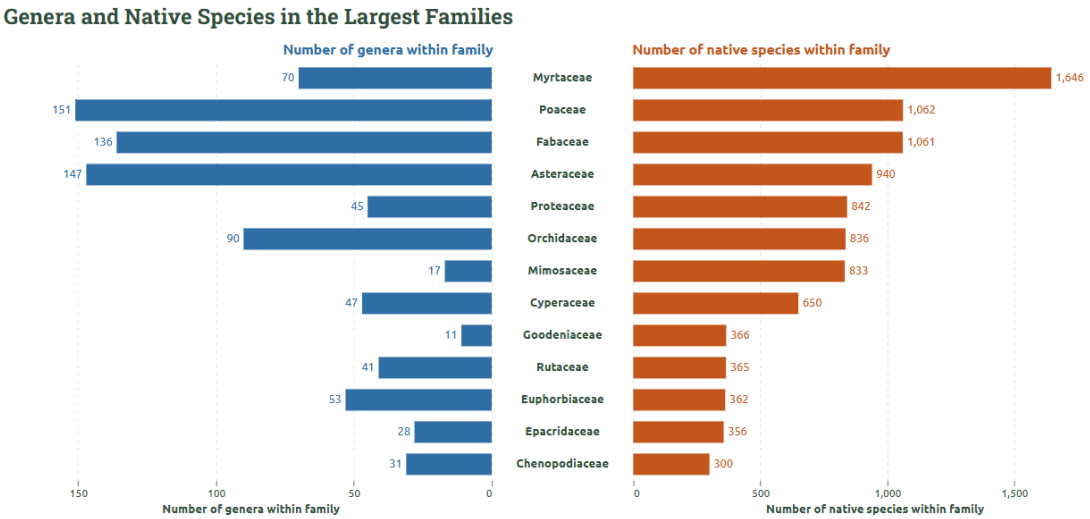
Why?

The graph shows how Australia's land is divided between different vegetation types, using area data from ABNG. The main task is to observe part to whole proportions, which is how much of the country each vegetation group covers. Viewers can compare the size of the groups to identify extremes and the provided legend as well as accompanied annotations aid understanding.

How?

A waffle chart is used because it presents parts of a whole, especially with the data that is provided in percentages and add up to 100%. The chart is a grid of 1000 square marks, where each square represents 0.1% of Australia, about 7,700 sqkm. The vegetation groups are categorical, and each is assigned a colour hue while the number of squares of each colour is a quantitative attribute that encodes the group's share of the land.

Chart 4 – Butterfly Chart



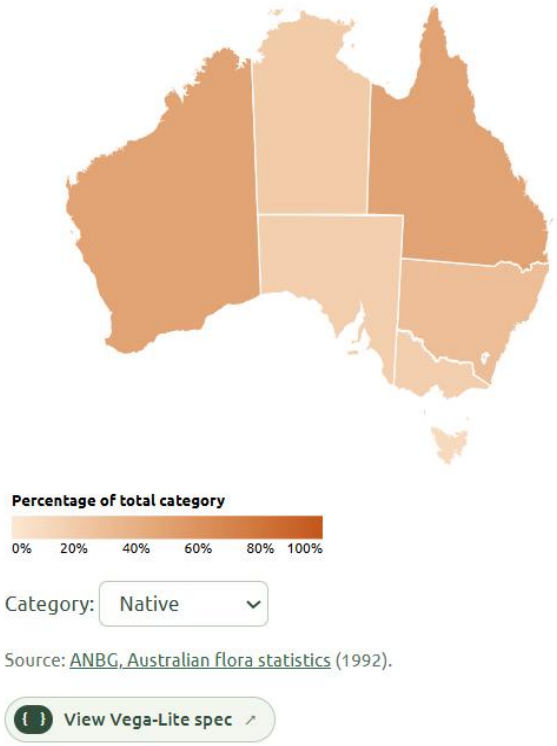
Why?

The graph shows how Australia's native flora is spread within its 13 largest plant families. For each family, it shows the number of genera and the number of native species. The main tasks of the graph are to compare the two values within each family and across families, and to identify outliers where the values differ greatly. Viewers can also rank families by size and discover which groups dominate Australia's flora, or which family houses the most or least number of native species.

How?

A butterfly chart is used because it puts two quantitative measures for the same categories side by side, mirrored around a shared axis, the family names which is a categorical attribute. The family names also sit in a central column and set the vertical position of each row so both measures for a family are read along the same line. Number of genera and native species are shown by the length of blue and red bars extending to the left and right, respectively. Bar length is the size channel that signals ranking while colour hue separates the two measures categorically.

Chart 5 – Choropleth map



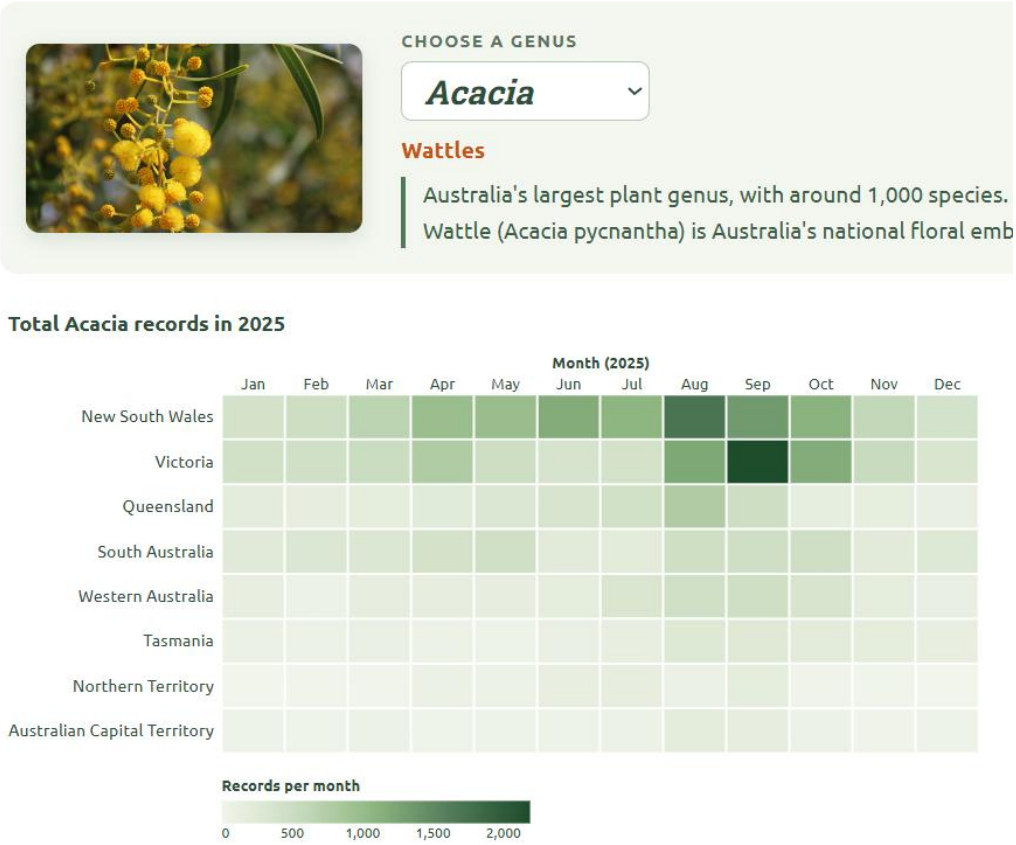
Why?

The graph aims to communicate the geographical distribution of native and naturalized flora across Australian states and territories. It allows the public to identify regional differences in plant composition and understand how the proportion of native and naturalised plants vary across the country.

How?

The visualisation uses an interactive choropleth map, where states are represented as geographical regions and encoded using colour saturation. The area marks represent states while colour intensity encodes the percentage of the selected flora category, sequentially increasing or decreasing towards a higher or lower value. An interactive dropdown menu allows users to switch between native and naturalised flora without changing the geographical layout, making differences in plant composition easy to identify.

Chart 6 – Heatmap



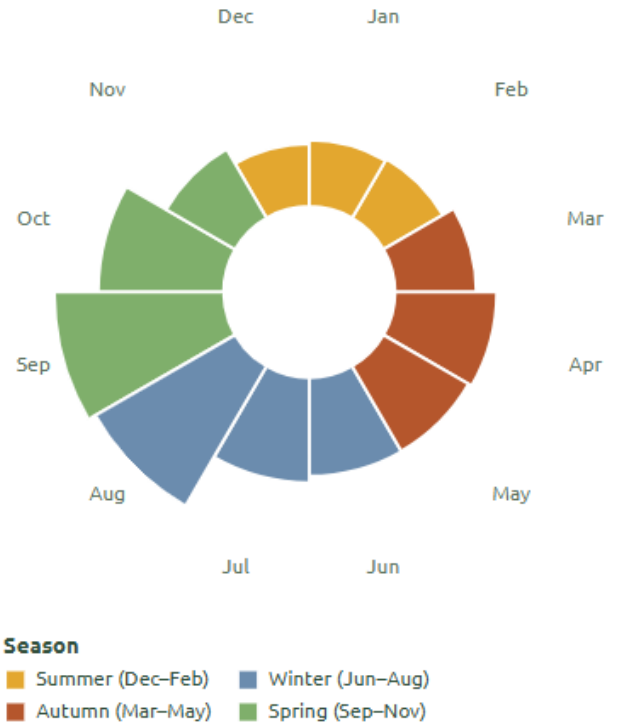
Why?

The heatmap aims to illustrate the temporal and geographical distribution of plant occurrence records Australia in 2025, allowing users to investigate how recorded observations of different plant genera vary by month and state, revealing seasonal and regional patterns

How?

The interactive nature of the heatmap uses rectangular marks positioned by month and state, with a sequential green colour scale encoding the number of occurrence records. A dropdown selector allows users to explore different genera, while accompanying images and descriptions provide botanical context, allowing for identification of patterns across both dimensions simultaneously.

Chart 7 – Radial chart



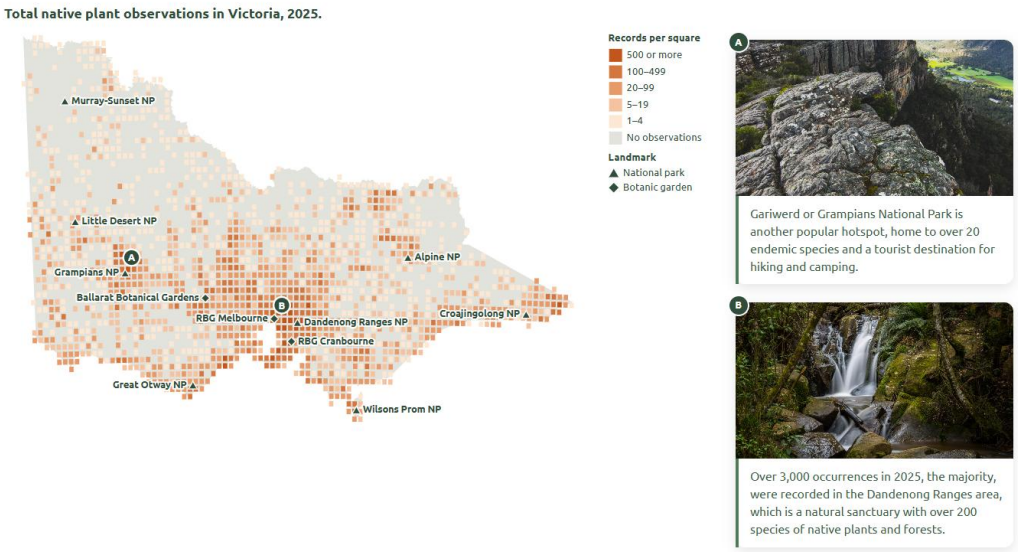
Why?

The chart compliments the heatmap by providing a seasonal overview of the occurrence records throughout the year. While the heatmap highlights differences across months and states, the radial chart emphasises seasonal patterns and makes it easier to identify periods with higher or lower observations.

How?

The visualisation uses a radial bar chart, arranged in a circular layout to represent the 12 months of the year. The marks are the arc shaped bars that represent individual months while angular position channels encode the month and radial length encodes the number of occurrence records. Colour hue distinguishes the four seasons, accompanied with legend. Its circular layout emphasises the annual seasonal cycle.

Chart 8 – Bin map



Why?

The map intends to illustrate the spatial distribution and concentration of native plant occurrence records across Victoria in 2025 and enable users to identify and compare geographical concentrations and patterns, as well as locate notable landmarks and discover areas of ecological interest.

How?

A density bin map uses square marks positioned geographically, while a sequential orange colour scale encodes the number of native plant occurrence records per cell. Distinct symbols identify national parks and botanical gardens, while annotations, images and descriptions provide additional ecological context. The idiom was chosen to reveal localised observation hotspot and spatial patterns that may be overlooked state level maps.

Chart 9 – UpSet plot



Source: Atlas of Living Australia

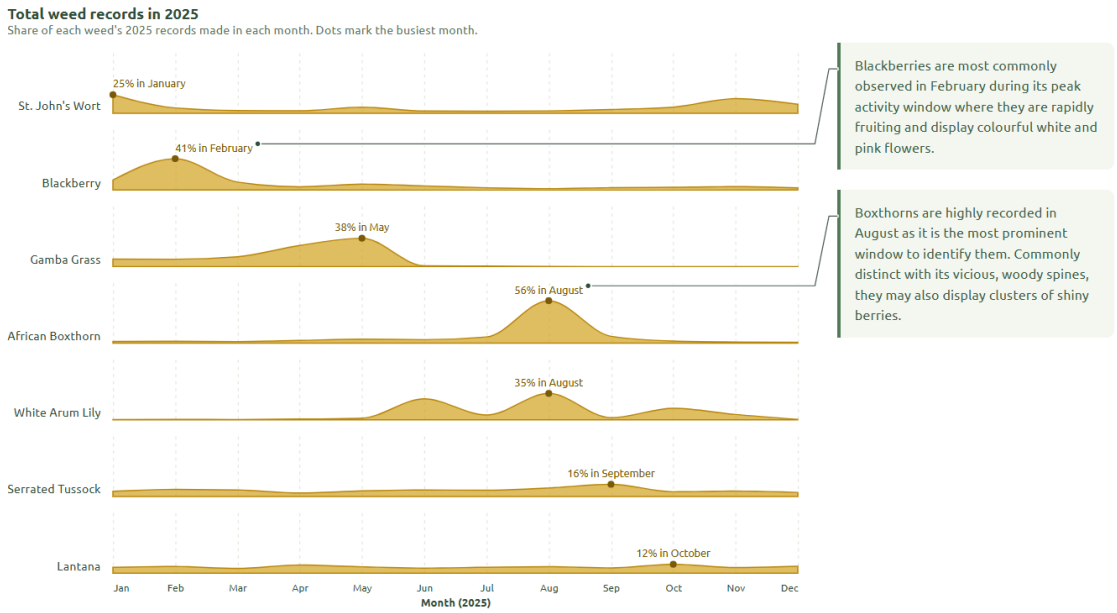
Why?

The visualisation enables users to compare the recorded occurrences of common Australian weeds, identify their inclusion in EPBC act classification lists, and explore the relationships between weed prevalence and classification.

How?

An UpSet plot, which is a connected dot matrix and a horizontal bar chart combo, uses circular marks and connecting lines to represent weed classification across the weed list, with colour distinguishing inclusion and exclusion. An aligned bar chart uses bar length to encode total occurrence records, allowing quantitative comparison between species. Accompanying images and annotations highlight notable invasive weeds, their environmental impacts and other important visual connotations.

Chart 10 – Ridgeline chart



Note: Hover over a ridge for details.

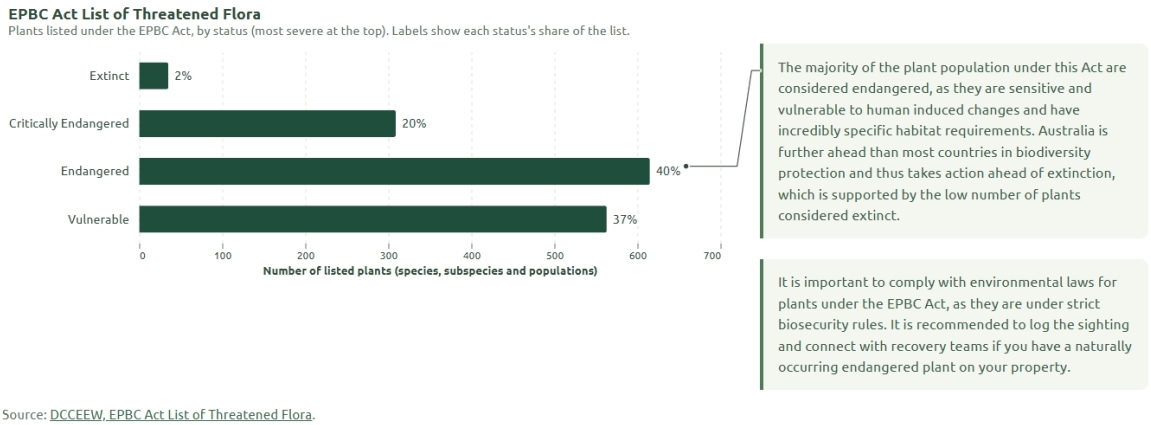
Why?

The chart allows for comparison of monthly weed occurrence patterns, identify peak observation periods for different species, and discover seasonal variations in the ALA provided recorded weed observations.

How?

A ridgeline plot uses vertically stacked area marks to represent the monthly distribution of occurrence records for each weed species. Horizontal position channels encode month, while ridge height represents the percentage of each species' annual records. Point marks and annotations highlight peak observation months, and interactive tooltips provide additional details

Chart 11 – Horizontal bar chart



Source: DCCEEW, EPBC Act List of Threatened Flora.

Why?

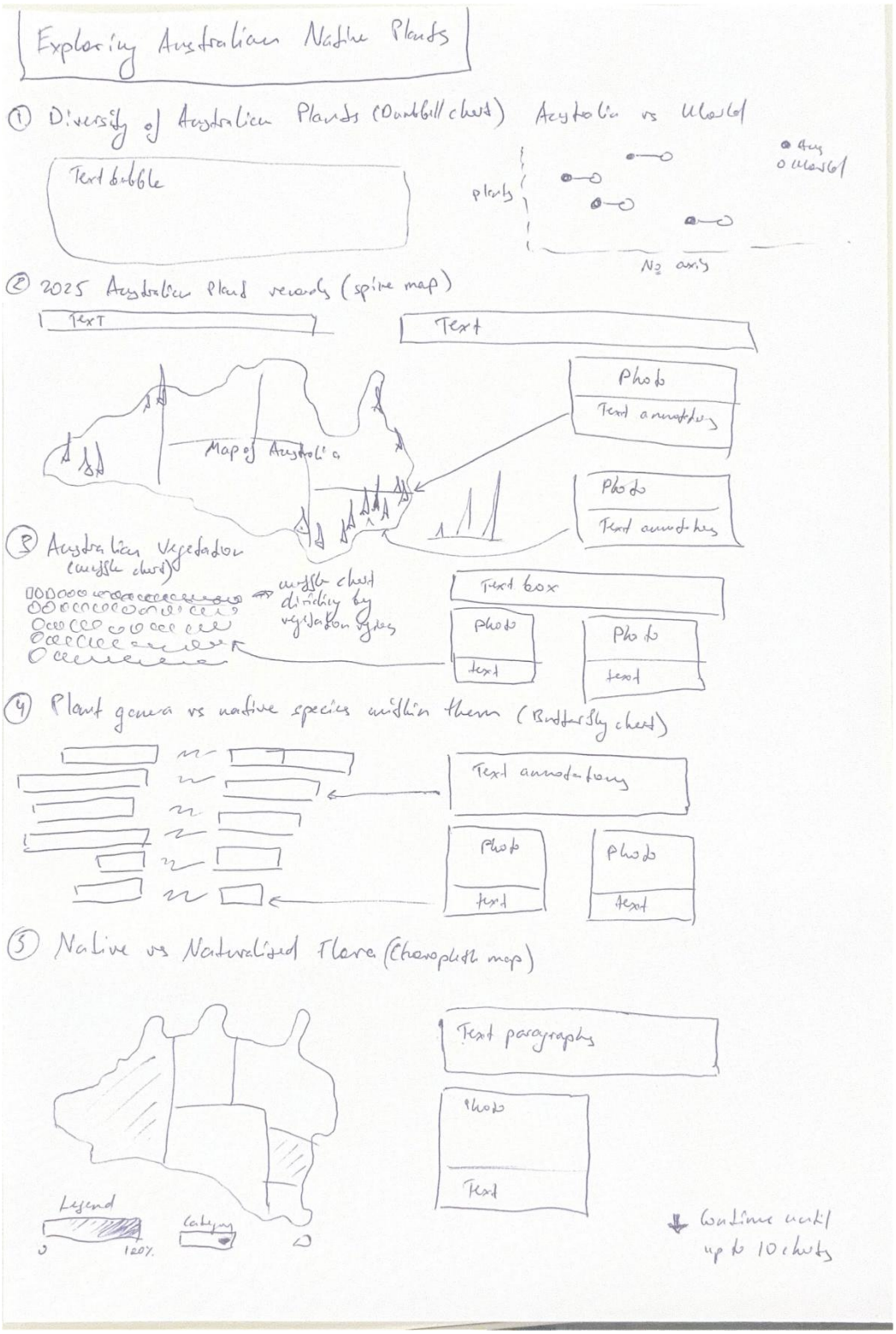
The visualisation allows users to compare the number and proportion of Australian flora listed under different EPBC act conservation statuses, identify the most common classifications and understand the extent of threatened plant diversity.

How?

A horizontal bar chart uses rectangular marks, with bar length encoding the number of listed plants and vertical position representing conservation status. Categories are ordered by severity, while percentage labels indicate each category's share of the total. Accompanying annotations provide conservation context.

Part C: Sketch

A URL to a PDF document with an image of your sketch on your GitHub account.
Tip: Test your URL in incognito mode to ensure it is a working URL



Part D: Full-size screenshot of your DV II

Exploring Australian Native Plants

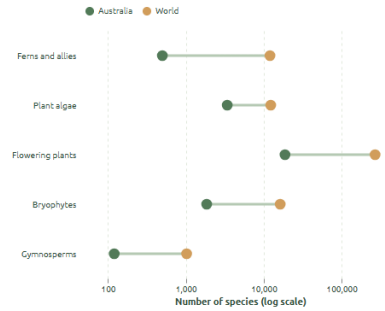
1 Diversity of Australian Plants

Australia has over **20,000** native vascular plant species, roughly 10% of the global total, and almost **9 in 10** of them (88%) are **endemic**: found nowhere else on Earth. Because the continent was geographically isolated for over 30 million years after its seperation from the Gondwana plate, its flora was able to evolve in complete independence, mostly with the ability to survive in nutrient-poor soils and extreme fire and drought regions that describe the climate of the country.

Flowering plants (angiosperms) make up the vast majority of the continent's flora. Smaller groups include **gymnosperms** (conifers, cycads and their relatives) and **bryophytes** (mosses, liverworts and hornworts), small spore-producing plants that help stabilise soil.

Australia holds about **7%** of the world's described flowering plants, but more than a quarter of it is described as plant algae, which are a diverse group of plant-like photosynthetic organisms that live primarily in aquatic environment. Aside from Australia, Brazil, China and New Zealand are also on the list with the highest number of endemic plants.

Plant Groups in Australia vs Worldwide



Note: The axis uses a log scale, so each gridline is ten times the one before.

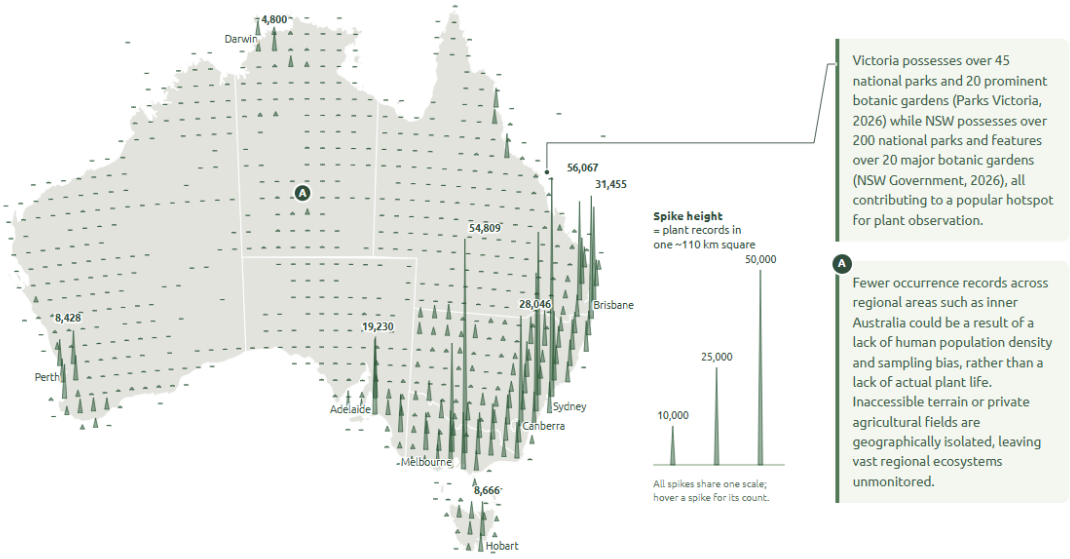
Source: DCCEEW, Numbers of Living Species in Australia and the World (2009).

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2 Plant observations in 2025

The spike map below shows the total recorded plant observations across Australia in 2025. From over **900,000** plant occurrences, the majority of them were recorded in the eastern states and along the coast, with the tallest spikes concentrated in Sydney and Melbourne.

Although a correlation can be drawn from recorded observations to plant growth density, it is important to consider that the data has a heavy "urban bias", specifically as cities have the highest density of human population meaning more concentration of citizen scientists to log sightings, as compared to regional areas.



Victoria possesses over 45 national parks and 20 prominent botanic gardens (Parks Victoria, 2026) while NSW possesses over 200 national parks and features over 20 major botanic gardens (NSW Government, 2026), all contributing to a popular hotspot for plant observation.

A

Fewer occurrence records across regional areas such as inner Australia could be a result of a lack of human population density and sampling bias, rather than a lack of actual plant life. Inaccessible terrain or private agricultural fields are geographically isolated, leaving vast regional ecosystems unmonitored.

Note: Each spike stands on a square about 110 km across; its height is proportional to the number of plant records in that square.

Source: Atlas of Living Australia (2025).

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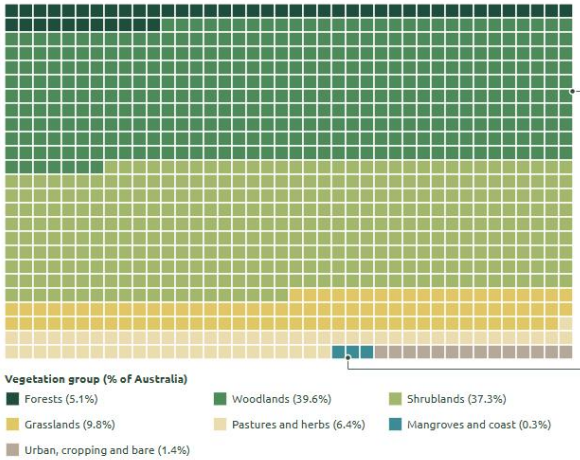
3 Australia's vegetation divide

Native vegetation is crucial for nourishing soils, sheltering and sustaining wildlife, protecting streams, wetlands and coastlines, as well as for cultivating our diverse flora (DCCEEW, 2021). Destruction and loss of

As observed from the waffle chart, Australia is heavily dominated by woodlands and open forests, with tall and low shrubs following second. Prominent botanical families such as **acacia** and **eucalyptus**, which all

native vegetation is a primary cause of land degradation, declining water quality, and thus the biggest cause of biodiversity loss. Although urban development and land clearing for agriculture are a big reason for this, weeds, fire spread and insect attacks are all equal threats.

thrive in wet and dry forests and shrublands thus dominate the landscape as a result of the vegetation divide.





Woodlands are comprised of widely spaced trees that are characteristically open and filled with sunlight, allowing for rich and diverse growth of grasses and wildflowers that thrive on sunlit ground.



Although possessing a small percentage of the vegetation divide, Australia is home to the third largest area of **mangroves** in the world, strictly inhabiting coastal environments such as tidal rivers and salt marshes where fine muddy sediments accumulate.

Note: Square counts are calculated from each group's area and rounded so they add to exactly 1,000. Each square = 0.1% of Australia (about 7,700 sq km).

Source: ANBG, Australian flora statistics, (1990).

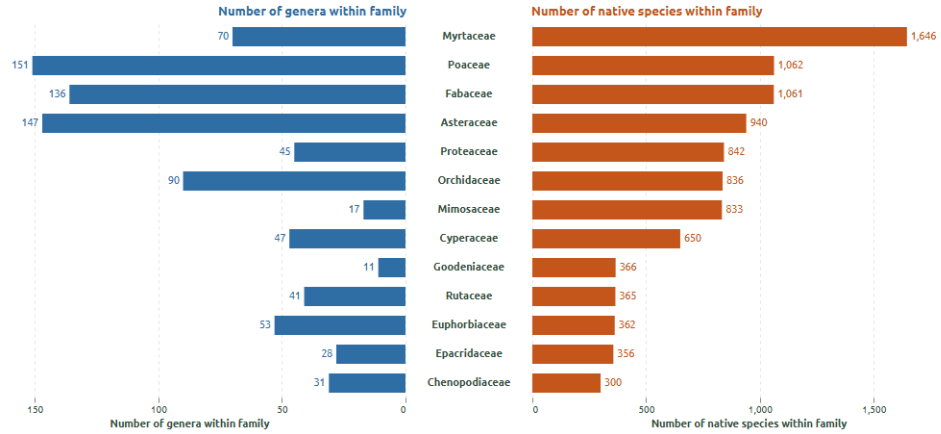
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4 Australia's Largest Plant Families

The **Myrtaceae** or myrtle family is Australia's most dominant and ecologically significant plant group, containing over **1600** native species that are unique to Australia only. Some examples of species include Eucalyptus, bottlebrushes and lemon myrtles. They form the primary structural vegetation across the continent, and are majorly concentrated around the wetter rainforests.

Although it contains the largest number of genera, the **Poaceae** or grass family is relatively lacking in unique native species. This family occupies the ground layer, sprawling across vast arid and savanna regions.

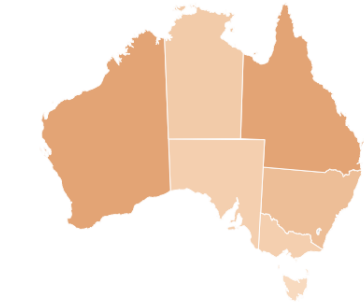
Genera and Native Species in the Largest Families



Source: ANBG, Australian flora statistics (1992).

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5 Native and Naturalised Plants



Percentage of total category

0% 20% 40% 60% 80% 100%

Category: Native

Source: ANBG, Australian flora statistics (1992).

View Vega-Lite spec

Queensland and Western Australia each contain almost half (**48%**) of Australia's native species. Tasmania has the smallest share, at **10%**. The correlation to the data can be made in relation to each state's vegetation share and size, where QLD and WA occupy a massive amount of regional space and spans tropical rainforests, savannas, arid deserts, and deserts, allowing for a rich cultivation of unique plants. Moreover, the isolated nature of a region like WA prevent naturalised species from spreading.

Switch to naturalised plants (species introduced from overseas that now grow wild) and the pattern moves south-east where New South Wales holds **64%** of them and Queensland **59%**, while the Northern Territory has only **13%**. This is directly tied to being the epicentre of early European settlement. For over 2 centuries, these areas experienced heavy maritime trade, agricultural clearing, and the deliberate introduction of European pasture grasses, crops, and ornamental garden plants, many of which spread into the wild. NT on the other hand has the lowest percentage of naturalised flora because of its remote geography, lower human population density, and extreme climate cycles that make it harder for typical European plants to survive and establish self sustaining wild populations without human intervention.

6 Common native plants and when and where do they grow?

This section covers the 10 most common native plants in Australia and tracks their growth over the 2025 year across seasons and states, drawn from occurrence records from ALA. Almost all of the plants are observed the most in spring, with autumn following second. An explanation for this trend is the rising temperatures, comfortable soil moisture, longer daylight hours and pollinator activity coinciding with the return of migratory birds and insects, all contributing to a favourable environment for plants to bloom.

Winter and summer are less popular in comparison, more likely due to less human activity for observation, as it coincides with fewer people out in the nature during the cold or abroad or busy during summer seasons. Environmentally it is less popular because of the dry soil due to the intense summer heat or cold soil and freezing temperatures during the cold. However, a small number of native Australian plants such as the **Grevilleas** and **Banksia** have evolved to endure Australia specific weather, which can be observed by their popularity all year round.



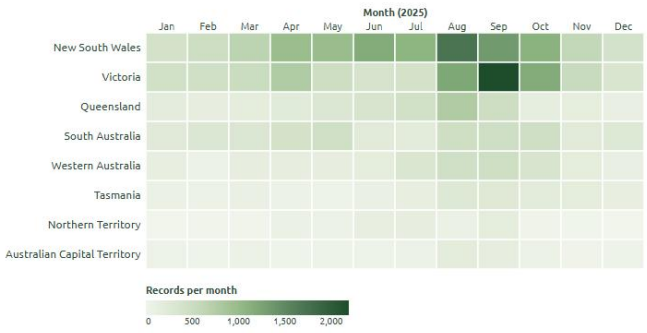
CHOOSE A GENUS

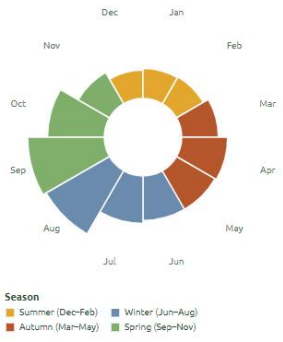
Acacia

Wattles

Australia's largest plant genus, with around 1,000 species. The Golden Wattle (Acacia pycnantha) is Australia's national floral emblem.

Total Acacia records in 2025





Note: Each wedge's area (beyond the centre hole) is proportional to that month's records.

Source: Atlas of Living Australia (2025).

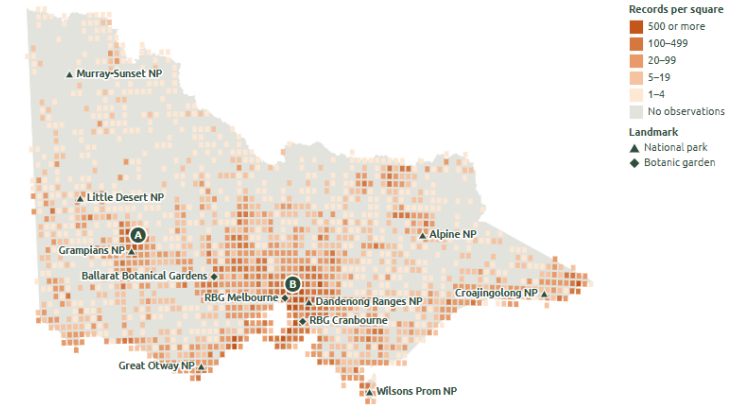
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7 Exploring Victoria

Vegetation in the Victorian region include forests, shrubs, grassy woodlands and heavy alpine regions and coastlines. The state is a popular hotspot for flowering plants, climbers and shrubs that seek shade and ground cover.

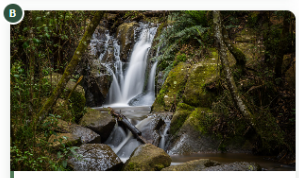
With a large reserve of national parks and botanical gardens, it offers a wealth of opportunities for citizens to observe pristine ecosystems and hiking trails.

Total native plant observations in Victoria, 2025.





Gariwerd or Grampians National Park is another popular hotspot, home to over 20 endemic species and a tourist destination for hiking and camping.



Over 3,000 occurrences in 2025, the majority, were recorded in the Dandenong Ranges area, which is a natural sanctuary with over 200 species of native plants and forests.

Note: Each square is about 10 km across, so every square covers the same area. Symbols mark approximate park and garden centres.

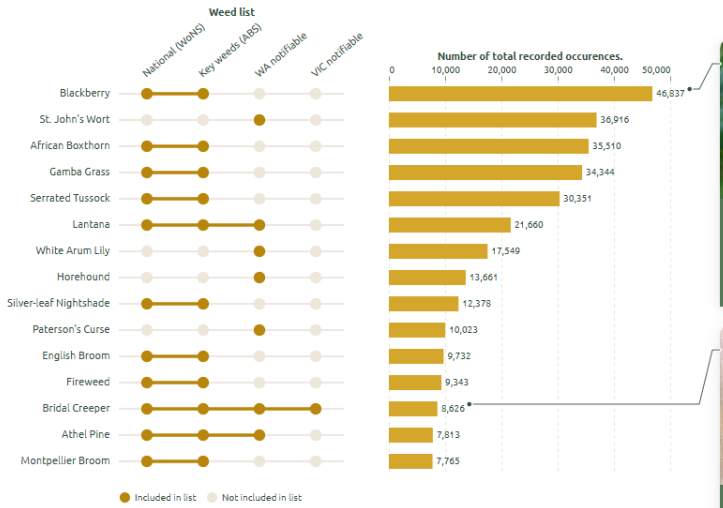
Source: iNaturalist (2025).

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8 Widespread Weeds

Weeds are a critical topic in Australian flora due to their invasive nature and threat to biodiversity and destruction of ecosystems. They reduce crop yields, threaten livestock, suffocate native plants and ultimately destroy wildlife habitats by destroying their food sources.

Australia has a number of identifiable lists to help its residents manage and handle weeds, of which the 15 most common are shown below, and their total number of occurrences across the country. Plants on the National Established Weed Priorities (WoNS) are considered especially invasive, commonly including **blackberries** and **boxthorns**. State specific weed lists legally regulate at the state and territory level and operate under their own biosecurity laws.





Blackberry is the most recorded weed and is consequently the most destructive weed in Australia, spreading across more than 8.8 million hectares and wielding a massive seed production amount of over 10,000 seeds per year.



Bridal Creeper is also considered the most aggressive as it is listed on every weed list. It is an introduced species from South Africa that has a vigorous climbing ability, dominating native woodlands, coastlines and riparian zones.

Source: Atlas of Living Australia

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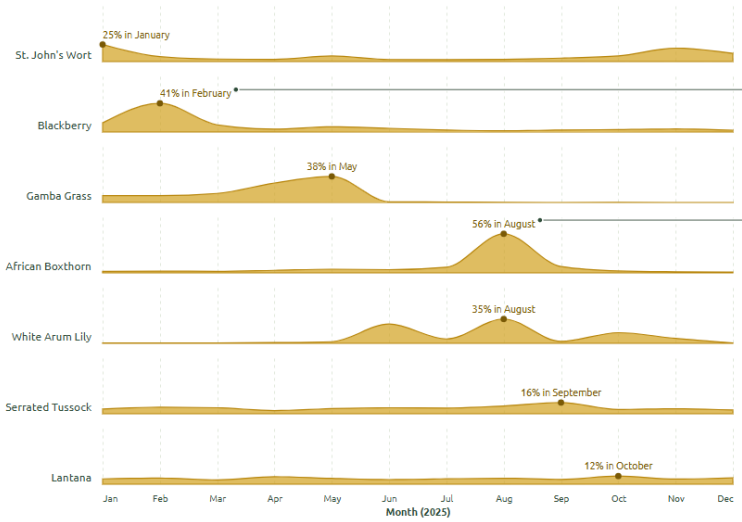
9 Weed growth across the year

Most weeds follow a typical annual lifecycle where they germinate during autumn and actively grow in winter, which is supported by the ridgeline chart below that details weed records during the year of 2025. Weed management should take priority during the autumn season while weeds are in their dormant germination window, to prevent flowering and vegetation in the later months.

Examples of action include applying herbicides to prevent seeds spreading, and hand pulling small patches or applying targeted sprays before they have a chance to bloom. After flowering, the government recommends immediate mowing or slashing to cut down seed heads before they mature and drop, to prevent infestation and multiplication the following year.

Total weed records in 2025

Share of each weed's 2025 records made in each month. Dots mark the busiest month.



Blackberries are most commonly observed in February during its peak activity window where they are rapidly fruiting and display colourful white and pink flowers.

Boxthorns are highly recorded in August as it is the most prominent window to identify them. Commonly distinct with its vicious, woody spines, they may also display clusters of shiny berries.

Note: Hover over a ridge for details.

Source: Atlas of Living Australia (2025).

View Vega-Lite spec

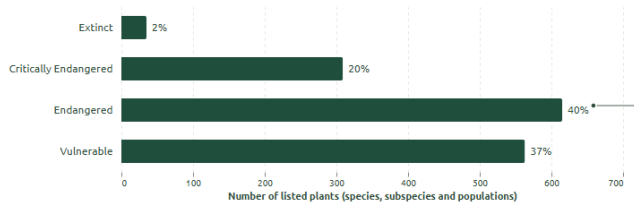
10 Plants under threat

The **Environment Protection and Biodiversity Conservation (EPBC) Act** is a central piece of national environmental legislation that maintains critical statutory lists to protect Australian biodiversity (DCCEEW, 2026). Over 2000 native plants are currently recognised on the list, with 4 distinct categories.

Extinct plants have no surviving members of their species; endangered plants are either facing a very high risk of extinction in the near future while vulnerable plants face a high risk of extinction in the medium term future. Key threats to these plants include invasive species, weeds and human or climate degradation.

EPBC Act List of Threatened Flora

Plants listed under the EPBC Act, by status (most severe at the top). Labels show each status's share of the list.



The majority of the plant population under this Act are considered endangered, as they are sensitive and vulnerable to human induced changes and have incredibly specific habitat requirements. Australia is further ahead than most countries in biodiversity protection and thus takes action ahead of extinction, which is supported by the low number of plants considered extinct.

It is important to comply with environmental laws for plants under the EPBC Act, as they are under strict biosecurity rules. It is recommended to log the sighting and connect with recovery teams if you have a naturally occurring endangered plant on your property.

Source: DCCEEW, EPBC Act List of Threatened Flora.

View Vega-Lite spec

FIT3179 Data Visualisation 2 · Created by Helen Zoljargal

Acknowledgement of AI

I used AI responsibly to help sift through data, improve grammar and coding for the website.
All output was then cross checked by me and validated before using for the project.

Sources

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- Parks Victoria (2026). What we manage.
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- Woodworkers Institute: The Charms of Acacia Wood (*Acacia* photo)

Photos

- weeds.org.au: Blackberry
- weeds.org.au: Bridal Creeper
- Hiking Fiasco: Olinda Falls, Dandenong Ranges National Park
- iNaturalist
- Wikipedia / Wikimedia Commons

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